

Understanding the Menstrual Cycle

A Complete Educational Guide
For students aged 10–18 · Parents · Educators

■ 28-day cycle	■ 4 phases	■ 4 key hormones	■ Ages 9–16 onset
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01 WHAT IS THE MENSTRUAL CYCLE?

The menstrual cycle is a natural, monthly biological process that occurs in the female body. It is the body's way of preparing for a possible pregnancy each month. If pregnancy does not occur, the cycle resets and begins again.

The cycle is counted from the first day of one period to the first day of the next. A typical cycle lasts between 21 and 35 days, with 28 days being the average. However, every person's cycle is unique — and variation is completely normal.

Key Facts at a Glance

Average cycle length	28 days (21–35 days is normal)
Period duration	3–7 days (average 5 days)
Age of first period	Between 9 and 16 years old
Blood lost per period	2–3 tablespoons (30–40 ml)
Number of phases	4 phases (Menstrual, Follicular, Ovulation, Luteal)
Key hormones involved	Estrogen, Progesterone, FSH, LH

■ Did You Know?

The word 'menstruation' comes from the Latin word 'mensis' meaning month, and the Greek word 'mene' meaning moon — because early humans noticed the cycle was roughly the same length as a lunar month.

The Four Phases — Overview

Phase	Days	Key Event	Dominant Hormone
■ Menstrual	1–5	Uterine lining sheds	Low estrogen & progesterone
■ Follicular	6–13	Follicles develop	Rising estrogen (FSH)
■ Ovulation	~14	Egg released	Peak estrogen + LH surge
■ Luteal	15–28	Body prepares for egg	Rising progesterone

02 THE FOUR PHASES — IN DETAIL

Phase 1 — MENSTRUAL PHASE · Days 1–5 · "The Reset"

What Happens

The menstrual phase marks the beginning of the cycle. The uterine lining — which built up during the previous cycle in preparation for a fertilised egg — is no longer needed and begins to shed. This shedding is what we experience as a period.

Hormones During This Phase

Both estrogen and progesterone are at their lowest levels during this phase. This hormonal drop is the biological trigger that initiates the shedding of the uterine lining.

Common Symptoms	Self-Care Tips
• Menstrual bleeding — light to heavy flow	✓ Apply heat (hot water bottle) to the lower abdomen to ease cramps
• Cramping caused by uterine muscle contractions	✓ Rest when you need to — your body is working hard
• Lower back pain and bloating	✓ Stay hydrated and eat iron-rich foods (spinach, lentils, red meat)
• Fatigue and low energy	✓ Light gentle movement such as walking or yoga can help
• Mood changes — irritability or emotional sensitivity	✓ Change pads or tampons every 4–6 hours

Phase 2 — FOLLICULAR PHASE · Days 6–13 · "The Build-Up"

What Happens

As the period ends, the pituitary gland releases FSH — follicle-stimulating hormone. This triggers the ovaries to begin developing small, fluid-filled sacs called follicles, each containing an immature egg. Over several days, one follicle becomes dominant and begins producing estrogen, which causes the uterine lining to thicken and rebuild.

Hormones During This Phase

FSH rises to stimulate follicle growth. Estrogen climbs steadily throughout the phase, peaking just before ovulation. Progesterone remains low.

Common Symptoms	Self-Care Tips
• Increased energy, motivation, and mental clarity	✓ This is a great time for social activities, new projects, and exercise
• Improved mood and sense of wellbeing	✓ Estrogen boosts serotonin — your mood is naturally elevated
• Clearer skin due to rising estrogen	✓ Eat foods rich in zinc and B vitamins to support follicle development
• Increased sociability and confidence	✓ Stay active — your energy is at its highest
• Cervical mucus becomes more fluid and clear	

Phase 3 — OVULATION · Around Day 14 · "The Main Event"

What Happens

Ovulation is the release of a mature egg from the dominant follicle in the ovary. It is triggered by a sharp surge in LH — luteinizing hormone. The egg travels down the fallopian tube toward the uterus. It survives for 12 to 24 hours. Because sperm can survive up to 5 days in the body, the fertile window is approximately the 5 days before ovulation plus the day of ovulation itself.

Hormones During This Phase

Estrogen peaks sharply just before ovulation, triggering the LH surge. LH spikes dramatically — this is the direct signal for the follicle to release the egg. Progesterone begins to rise after ovulation.

Common Symptoms	Self-Care Tips
• Peak energy and libido	✓ If tracking fertility, this is your peak fertile window
• Cervical mucus is clear, slippery, and stretchy (like egg whites)	✓ The LH surge can be detected using ovulation predictor test strips
• Mild one-sided pelvic pain known as mittelschmerz	✓ Mittelschmerz (ovulation pain) is normal and temporary
• Slight rise in basal body temperature (0.2–0.5°C)	✓ Basal body temperature rises slightly after ovulation — useful for tracking
• Heightened senses — sharper smell, clearer vision	

Phase 4 — LUTEAL PHASE · Days 15–28 · "The Preparation"

What Happens

After ovulation, the empty follicle transforms into the corpus luteum — a temporary gland that produces progesterone. Progesterone thickens the uterine lining further, making it spongy and nourishing, ready to receive a fertilised egg. If fertilisation does not occur, the corpus luteum breaks down after 10–12 days. Progesterone and estrogen both drop, causing the lining to detach — triggering the next period.

Hormones During This Phase

Progesterone dominates this phase and reaches its peak around day 21. Estrogen has a secondary smaller rise before both hormones fall toward the end of the phase. The fall of both hormones triggers PMS symptoms and then menstruation.

Common Symptoms	Self-Care Tips
• Bloating and breast tenderness	✓ Reduce salt intake to minimise bloating
• Mood changes — irritability, anxiety, or low mood (PMS)	✓ Limit caffeine and alcohol which worsen PMS symptoms
• Food cravings, especially sweets and carbohydrates	✓ Dark chocolate and magnesium-rich foods can reduce cramps
• Fatigue and disrupted sleep	✓ Prioritise sleep — progesterone can cause drowsiness
• Headaches and mild cramps toward the end of the phase	✓ Gentle exercise and stress management improve mood

03 KEY HORMONES EXPLAINED

Estrogen

Produced by: Ovaries (mainly)

The primary female sex hormone. Drives the thickening of the uterine lining during the follicular phase. Boosts mood, energy, and skin clarity. Peaks just before ovulation.

Progesterone

Produced by: Corpus luteum (after ovulation)

Maintains the thickened uterine lining during the luteal phase. Prepares the body for potential pregnancy. When it drops, menstruation begins. Associated with PMS symptoms.

FSH — Follicle-Stimulating Hormone

Produced by: Pituitary gland (in the brain)

Released at the start of the follicular phase. Signals the ovaries to begin developing follicles. Each follicle contains an immature egg. One follicle becomes dominant.

LH — Luteinizing Hormone

Produced by: Pituitary gland (in the brain)

The LH surge is the direct trigger for ovulation. A sharp spike in LH causes the dominant follicle to burst open and release the mature egg. Home ovulation test kits detect this surge.

■ How Hormones Work Together

Think of the four hormones as a relay team. FSH starts the race by growing follicles. Estrogen builds momentum — thickening the lining and peaking before the handoff. LH fires the starting gun for ovulation. Then Progesterone takes over, maintaining the lining until the race either ends (period) or continues (pregnancy).

04 HORMONE LEVELS ACROSS THE CYCLE

The chart below shows approximate relative hormone levels during each phase of the cycle. Values are relative (not absolute) and represent typical patterns.

Phase	Estrogen	Progesterone	LH	FSH
■ Menstrual (1–5)	Very Low ▼	Very Low ▼	Low	Rising ↑
■ Follicular (6–13)	Rising ↑	Low	Low	High ↑
■ Ovulation (~14)	Peak ▲	Low→Rising	Surge ▲▲	Drops ↓
■ Luteal (15–28)	Moderate	Peak ▲	Low	Low
End of Luteal (27–28)	Drops ↓	Drops ↓	Low	Starts rising

■ Reading the Table

▲ = Peak level ↑ = Rising ↓ = Falling ▼ = Low ▲▲ = Surge (very sharp spike). The LH surge during ovulation is the sharpest hormonal change in the entire cycle — it can be detected with an ovulation predictor kit (OPK) available at pharmacies.

05 WHAT TO EXPECT — SYMPTOMS & SELF-CARE

Managing Period Pain (Dysmenorrhea)

Period cramps are caused by prostaglandins — hormone-like substances that trigger uterine contractions to shed the lining. Mild to moderate cramping is normal. Severe pain that prevents daily activity should be discussed with a doctor.

■ Heat therapy	A hot water bottle or heating pad on the lower abdomen relaxes muscle contractions.
■ Light exercise	Walking, yoga, or gentle stretching releases endorphins — natural pain relievers.
■ Hydration	Drinking water reduces bloating and can ease cramping.
■ Nutrition	Iron-rich foods (spinach, lentils), omega-3s (fish, walnuts), and magnesium (dark chocolate, nuts) all help.
■ Pain relief	Ibuprofen or paracetamol can help — always check with an adult before taking medication.
■ Rest	Your body is doing significant work. Adequate sleep and rest are important.

Understanding PMS (Premenstrual Syndrome)

PMS refers to physical and emotional symptoms that occur during the luteal phase, typically in the 7–14 days before a period. It is caused by the rise and then fall of progesterone and estrogen. PMS resolves once menstruation begins.

Physical Symptoms	Emotional Symptoms
Bloating and water retention	Mood swings
Breast tenderness	Irritability and short temper
Headaches	Anxiety or low mood
Fatigue and poor sleep	Tearfulness
Food cravings	Difficulty concentrating
Acne flare-ups	Feeling overwhelmed

06 TRACKING YOUR CYCLE

Tracking your menstrual cycle is one of the most useful things you can do for your health. It helps you understand your body, predict your period, manage symptoms proactively, and provide accurate information to a healthcare provider if something feels wrong.

What to Track

■ Period start date	Mark the first day of bleeding as Day 1 of your cycle.
■ Period end date	Note when bleeding stops to calculate period length.
■ Flow intensity	Light, medium, heavy, or very heavy — useful for spotting changes.
■ Symptoms	Cramps, headaches, bloating, breast tenderness, mood changes.
■ Sleep quality	Note if your sleep is disrupted — common in the luteal phase.
■ Energy levels	Track your energy across the month to see the natural rhythm.
■ Mood	Notice patterns in how your mood changes with your cycle.

How to Calculate Your Cycle Length

Count from Day 1 of one period (first day of bleeding) to Day 1 of the next period. That number is your cycle length. Track for 3–6 months to find your personal pattern. Most people find their cycle is consistent within a few days.

■ Example

Period starts March 1. Next period starts March 29. Cycle length = 28 days. Period starts again April 26. Cycle length = 28 days. Pattern confirmed.

07 COMMON MYTHS vs FACTS

✗ MYTH	Periods are dirty or impure.
✓ FACT	Menstrual blood is simply blood and tissue — exactly like any other blood. It is a healthy, natural process.
✗ MYTH	You can't swim or exercise during your period.
✓ FACT	You absolutely can. Swimming, sports, and exercise are all safe and often help reduce cramps by releasing endorphins.
✗ MYTH	All menstrual cycles are exactly 28 days.
✓ FACT	Cycles range from 21 to 35 days and still be completely normal. 28 days is just the average.
✗ MYTH	You lose a lot of blood during a period.
✓ FACT	The average total blood loss per period is only 2–3 tablespoons (30–40ml). It looks like more because it mixes with fluid.
✗ MYTH	Period pain is just in your head.
✓ FACT	Period cramps are caused by real muscle contractions triggered by prostaglandins. They are a genuine physical experience.
✗ MYTH	You can't get pregnant during your period.
✓ FACT	While unlikely, it is possible — especially with short cycles. Sperm can survive up to 5 days inside the body.
✗ MYTH	Irregular cycles always mean something is wrong.
✓ FACT	Irregular cycles are common in the first 1–3 years after periods begin, and can be caused by stress, weight changes, or illness.
✗ MYTH	Tampons affect virginity.
✓ FACT	Virginity is a social concept, not a medical one. Using a tampon does not change anything about your body.

08 WHEN TO SEE A DOCTOR

Most period experiences are normal. However, there are certain signs that warrant a conversation with a healthcare provider. Always trust your instincts — if something feels wrong, it is always worth getting checked.

Periods lasting longer than 7 days

May indicate hormonal imbalance or a condition such as fibroids.

Soaking through a pad or tampon every 1–2 hours

Heavy bleeding (menorrhagia) can lead to anaemia and needs assessment.

Severe pain that prevents daily activities

Could indicate endometriosis, PCOS, or other conditions — not something to push through.

No period by age 16

Primary amenorrhea should always be evaluated by a doctor.

Periods stopping for 3+ months (when not pregnant)

Secondary amenorrhea has many causes including stress, weight, or hormonal issues.

Cycles shorter than 21 days or longer than 35 days consistently

After the first 2–3 years of menstruation, a very irregular cycle warrants investigation.

Bleeding between periods (spotting) regularly

Occasional spotting can be normal but frequent spotting should be assessed.

Unusual discharge with odour or colour

May indicate infection — seek medical advice promptly.

Remember

A healthcare provider — whether a doctor, gynaecologist, or school nurse — is always your best resource for personal health questions. There is no such thing as a question that is too embarrassing to ask. They have heard it all before, and they are there to help you.

09 QUICK REFERENCE SUMMARY

	Menstrual Days 1–5	Follicular Days 6–13	Ovulation ~Day 14	Luteal Days 15–28
Main event	Lining sheds	Follicles grow	Egg released	Lining maintained
Estrogen	Very Low	Rising → High	Peak	Moderate
Progesterone	Very Low	Low	Starts rising	High → Drops
Energy	Low	Rising → High	Peak	Moderate → Low
Mood	Lower / sensitive	Positive / clear	Confident / social	Variable / PMS
Skin	May flare	Clear / glowing	Best	May flare
Best for	Rest & self-care	New projects	Social / active	Reflection / rest
Watch for	Cramps, fatigue	High energy	Mild pelvic pain	PMS symptoms

A NOTE TO THE READER

Understanding your menstrual cycle is one of the most powerful things you can do for yourself. It is not just about periods — it is about understanding your energy, your emotions, your skin, and your overall wellbeing across the entire month. Every body is different. Every cycle is valid. Variation is normal. You are not broken if your cycle does not match a textbook. If you have questions — ask. If something feels wrong — speak up. And if you simply want to understand yourself better — keep learning. You deserve that knowledge.

Period Education App · This guide is for educational purposes only · Not a substitute for medical advice